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Testing Heavy Neutral Leptons with Multi-Messenger Observations of NGC 1068

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The active galactic nucleus NGC 1068, recently identified as a high-energy neutrino source by IceCube, exhibits a GeV γ-ray flux that exceeds predictions from opaque corona cascade models. This tension suggests the existence of non-attenuated energy transport mechanisms from the compact core. We investigate whether heavy neutral leptons (HNLs), coupled to the Standard Model through mixing with muon neutrinos, can provide such a channel, and we derive the resulting *exclusionary constraint* on this HNL scenario from the multi-messenger observations of NGC 1068. We calculate the production of HNLs in charged-pion and muon decays associated with hadronic interactions, explicitly tracking the mass-dependent polarization redistribution. We then derive the spectrum from the visible $N \to \nu_\mu + e^+ + e^-$ decay channel and evaluate its secondary electromagnetic emission. We find that the predicted photon flux is at least 7 to 8 orders of magnitude below the observational upper limits once experimental bounds on $|U_{\mu N}|^2$ are applied, which excludes this HNL channel as the origin of the γ-ray excess. The polarization-tracking framework developed in this work is applicable not only to astrophysical studies but may also provide useful theoretical input for decay simulations, detection-efficiency analyses, and mass reconstruction in future accelerator-based HNL searches.

Keywords: HNL; Inverse Seesaw; multi-messenger; NGC 1068

1. Introduction

Multi-messenger astronomy has turned astrophysical accelerators into laboratories for physics beyond the Standard Model (BSM). The IceCube collaboration has identified the active galactic nucleus (AGN) NGC 1068 as a point source of neutrinos at energies around 1 TeV,[1] and the KM3NeT/ARCA telescope has reported a high-energy event candidate possibly reaching 220 PeV.[2] These observations indicate that hadronic acceleration operates in at least some AGN cores, and they constrain the neutrino flux that any transport mechanism in such a source must reproduce.

The same source is anomalous in γ-rays. The GeV γ-ray flux of NGC 1068

measured by Fermi-LAT exceeds the predictions of opaque corona cascade models, in which the γ rays are generated in the optically thick core and attenuated before escape.[3,4] The excess therefore suggests a transport channel that carries energy out of the compact core without being absorbed, and that converts this energy into observable photons only beyond the opaque region. Neutral, weakly interacting particles are plausible candidates for such a channel: they escape the core without electromagnetic attenuation and decay or scatter in the surrounding environment, where their secondaries radiate. The question we address is whether a specific BSM particle can play this role.

HNLs are motivated here on two levels. Theoretically, the non-zero but extremely small active neutrino masses require physics beyond the Standard Model.[5,6] HNLs are among the simplest mediators of neutrino mass and can simultaneously provide a dark-matter candidate and a target for laboratory searches.[7] In the Inverse Seesaw (ISS) mechanism the smallness of the light neutrino mass is controlled by a small lepton-number-violating parameter rather than by a very large Majorana mass, so that the mixing with active neutrinos can remain large enough to be phenomenologically relevant. Phenomenologically, a MeV-scale HNL mixed with the muon neutrino can be produced in the principal neutrino production chain of NGC 1068 (pion and muon decays) and possesses a visible decay channel, making it a plausible candidate for the transport channel of this source. Terrestrial experiments have not yet observed a definite HNL signal, but they constrain the mixing with the muon neutrino to $|U_{\mu N}|^2 \lesssim 10^{-4}$ for $3\,\mathrm{MeV} < m_N < 19.5\,\mathrm{MeV}$;[8] the bound is slightly weaker but of the same order over the 10 MeV to 30 MeV range considered here. These considerations motivate HNLs as a representative candidate; other neutral candidates, such as axion-like particles, have been studied separately for NGC 1068,[9] and we do not attempt a comparative survey here.

A dedicated calculation is therefore required. A quantitative statement cannot rely on the massless-neutrino approximation: for m_N in the 10 MeV to 30 MeV range the HNL is non-relativistic in the parent rest frame, phase-space suppression is significant, and the helicity produced in the weak decay is redistributed by the boost in a mass-dependent way. This is already visible at the production stage: as shown in Figs. (2)–(4) below, right-handed HNLs, which are absent in the massless limit, become an appreciable fraction of the pion-decay yield as m_N approaches the phase-space boundary, and Fig. (4) shows the same effect in muon decay. These polarization states are retained in the subsequent decay: the visible channel $N \to \nu_\mu + e^+ + e^-$ inherits the helicity through the angular distributions $g_e^{(i)}(x, \cos\theta)$ of Eqs. (24)–(25), so that the electron spectrum, and hence the γ-ray yield, depends on the polarization of the parent HNL. Polarization-resolved treatments of HNL production and decay have been developed for accelerator-based HNL searches,[10] but their application to an astrophysical source requires a source-specific production spectrum and environment. A polarization-resolved treatment is therefore adopted here to convert the NGC 1068 multi-messenger data into a quantitative statement about this channel, and it also makes the framework reusable for other sources and

other HNL mass windows.

In this paper we test whether this HNL channel can account for the γ-ray excess of NGC 1068, and we derive the corresponding exclusionary constraint on this scenario. We compute the HNL source spectrum from $\pi^{\pm} \to \mu^{\pm} + N$ and $\mu^{\pm} \to N + e^{\pm} + \bar{\nu}_e$, separating the two polarization states and boosting to the laboratory frame while tracking the polarization redistribution. We then derive the electron/positron spectrum from the visible decay $N \to \nu_\mu + e^+ + e^-$, propagate these leptons in a representative environment of NGC 1068, and compute the resulting electromagnetic emission with GAMERA.[11] Comparing with the observed γ-ray flux, we find that the predicted photon flux lies at least 7 to 8 orders of magnitude below the observational upper limits once the experimental bound on $|U_{\mu N}|^2$ is applied. The HNL channel is therefore excluded as the origin of the γ-ray excess.

The paper is organized as follows. Section 2 gives the HNL coupling, the ISS parameter range, the single-Dirac approximation, and the decay channels. Section 3 presents the polarization-resolved calculations of HNL production in pion and muon decays and of the visible three-body decay $N \to \nu_\mu + e^+ + e^-$, and collects them into the electron source function injected into the radiation calculation. Section 4 applies this source function to the environment of NGC 1068, compares with observations, and gives the suppression budget and its robustness. Section 5 summarizes our conclusions.

2. ISS Model: Parameters and Prerequisites

This section provides the four ingredients needed below: the coupling of HNLs to the SM, the allowed ranges of m_N and $|U_{\mu N}|^2$ in the ISS, the justification of the single-Dirac approximation, and the visible decay channel with its branching ratio. The production and decay calculations of Sec. 3 and the experimental bound and suppression budget of Sec. 4 rest on them.

2.1. *Coupling*

HNLs interact with the SM only through their mixing with the active neutrinos. Therefore any process that produces ν_μ also produces N, with a rate multiplied by $|U_{\mu N}|^2$, provided that it is kinematically allowed. Section 3 uses this to compute N production in charged-pion and muon decay chains.

2.2. *ISS and parameter ranges*

In the Type-I seesaw:[12–17]

$$m_\nu = \frac{m_D^2}{M_R},\ m_N = M_R,\ |U|^2 = \left(\frac{m_D}{M_R}\right)^2 \tag{1}$$

For an observable M_R the mixing is usually too small to be probed. The Inverse Seesaw (ISS)[12–17] introduces another fermion field S, and the mass matrix reads

$$\left(\overline{\nu_L}\ \overline{\nu_R^c}\ \overline{S^c}\right)\begin{pmatrix} 0 & m_D & 0 \\ m_D & 0 & M \\ 0 & M & \mu \end{pmatrix}\begin{pmatrix} \nu_L^c \\ \nu_R \\ S \end{pmatrix} \tag{2}$$

where μ is a small lepton-number-violating mass term. For $\mu \ll M$ and $\mu \ll m_D$,

$$m_\nu = \mu\frac{m_D^2}{M^2},\ m_N = M,\ |U|^2 = \frac{m_\nu}{\mu} \tag{3}$$

The key point is that the smallness of the light neutrino mass is controlled by the small parameter μ rather than by a very large M, so that for comparable neutrino masses the mixing can be larger than in the Type-I case.

For the parameters used here, $|U|^2$ can reach the 10^{-4} level: taking $m_\nu \sim 0.05$ eV and $\mu \sim$ keV gives $|U|^2 = m_\nu/\mu \sim 10^{-4}$, while $\mu \ll m_D, M$ is satisfied. This is comparable to the experimental bound quoted in Sec. 1 and motivates the benchmark mixing adopted below.

2.3. *Single-Dirac approximation*

For small μ the heavy states form a nearly degenerate pseudo-Dirac pair.[18,19] Over the range $m_N \simeq 10\,\mathrm{MeV}$ to $30\,\mathrm{MeV}$ and $\mu \ll m_D, M$ considered here, the mass splitting is much smaller than m_N, so that production and decay cannot resolve the two states; summing over helicities is therefore equivalent to a single Dirac fermion. The helicity-resolved calculation of Sec. 3, with $s = \pm 1$, rests on this approximation.

2.4. *Decay channels and visible branching ratio*

Because the HNLs considered here are lighter than the pion, their production and decay are governed by low-energy weak processes. They decay only into leptons: a three-neutrino channel (invisible) and a $\nu_\mu e^+ e^-$ channel (visible). Section 3.3 gives the explicit value $BR \simeq 0.11$, which enters the suppression budget of Sec. 4.3.

3. Polarization-Resolved HNL Production and Decay

Under AGN conditions, high-energy cosmic rays undergo $p-p$ and $p-\gamma$ interactions, which are thought to be the main sources of secondary particles;[20] these interactions produce copious pions, whose decays subsequently generate high-energy neutrinos and γ-rays.[21] We consider the two primary production channels for muon-flavored neutrinos, charged-pion decay and muon decay, and then the visible decay of the HNL; polarization is tracked through all three stages.

Throughout this section, E_N and E denote laboratory-frame energies of the HNL and of the electron, respectively, while x and y are the corresponding dimensionless variables in the parent rest frame, defined where they first appear. The helicity label $s = \pm 1$ denotes the HNL spin projection along its momentum, with $s = +1$ for a right-handed HNL; the same convention is used in all three stages below.

3.1. *Production in pion decay*

The decay $\pi^+ \to \mu^+ + N$ is the HNL analogue of the process $\pi^+ \to \mu^+ + \nu_\mu$.[22]

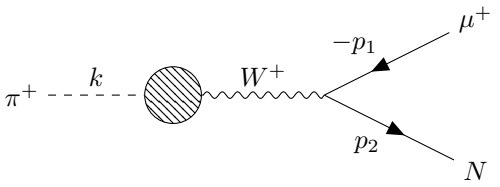


Fig. 1. Feynman diagram of π^+ decaying into μ^+ and massive HNLs.

By adapting the standard pion-decay calculation, we find that the decay width to HNL is suppressed by the mixing angle and modified by a kinematic factor:

$$\frac{\Gamma_{\pi\to\mu N}}{\Gamma_{\pi\to\mu\nu}} = |U_{\mu N}|^2 \times \rho_\pi(m_N) \tag{4}$$

where

$$\rho_\pi(m_N) = \frac{(m_\pi^2(m_\mu^2+m_N^2) - (m_\mu^2-m_N^2)^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)}}{m_\mu^2(m_\pi^2-m_\mu^2)^2} \tag{5}$$

$$\lambda(x,y,z) = x^2+y^2+z^2-2xy-2yz-2zx \tag{6}$$

The function ρ_π encodes the phase-space suppression.

The HNL polarization is important because it controls the angular dependence of the subsequent decay. In the pion rest frame we have

$$\Gamma_{(s)} \sim (m_\mu^2+m_N^2)m_\pi^2 - (m_\mu^2-m_N^2)^2 - s(m_\mu^2-m_N^2)\sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)} \tag{7}$$

where s is the polarization of HNL, and $s=+1$ for right-handed polarized HNLs.

Since the parent pion is a spin-zero scalar, the decay is isotropic in its rest frame. However, the polarization of the massive HNL needs to be carefully tracked, as the subsequent Lorentz boost to the laboratory frame induces a mass-dependent polarization redistribution. For a pion spectrum $J(E_\pi)$, we get

$$Q_N^{(s)}(E_N) = \int_{\lambda^-}^{\lambda^+} J(\frac{E_N}{x})|U_{\mu N}|^2\rho_\pi \times r_{(s)}(x,E_N) \times \frac{dx}{(\lambda^+-\lambda^-)\cdot x} \tag{8}$$

where

$$\lambda^\pm = \frac{m_\pi^2 - m_\mu^2 + m_N^2 \pm \sqrt{\lambda(m_\pi^2,m_\mu^2,m_N^2)}}{2m_\pi^2} \tag{9}$$

$$\cos\theta_r = \frac{x\,(m_\pi^2-m_\mu^2+m_N^2) - 2m_N^2}{x\sqrt{(1-\frac{m_N^2}{E_N^2})\,\lambda(m_\pi^2,m_\mu^2,m_N^2)}} \tag{10}$$

$$r_{(s)}(x,E_N) = \frac{\Gamma_{(s)}\frac{1+\cos\theta_r}{2} + \Gamma_{(-s)}\frac{1-\cos\theta_r}{2}}{\Gamma_{(s)}+\Gamma_{(-s)}} \tag{11}$$

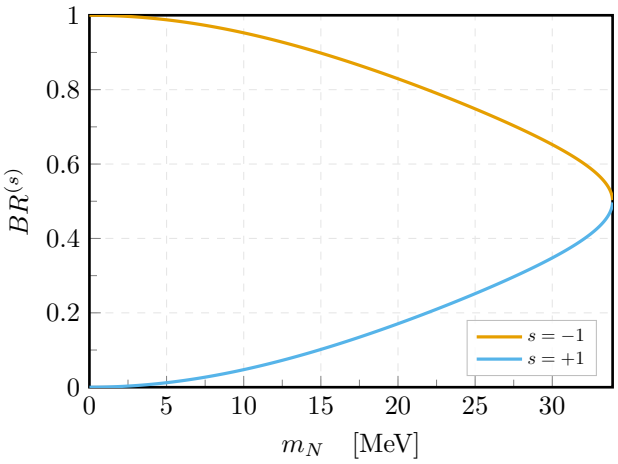


Fig. 2. Eq. (7) for different s. In pion decay, right-handed HNLs are more likely to be produced close to the phase space boundary.

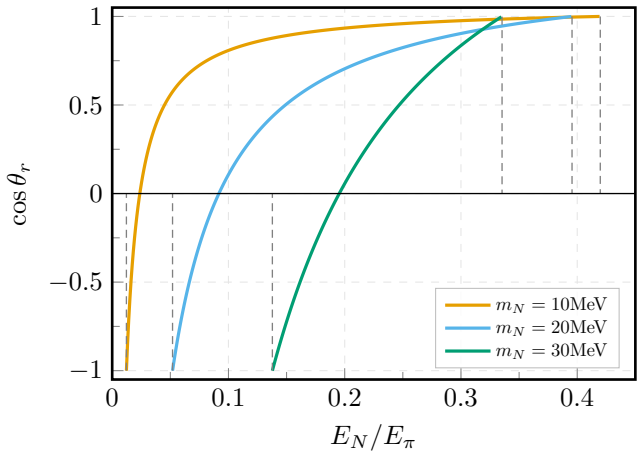


Fig. 3. Eq. (10) visualized with different m_N values. Eq. (10) determines how much helicity flip will happen during Lorentz boost. When m_N is small $\cos\theta_r \approx 1$, combined with Fig. (2), HNLs produced are almost left-handed; as m_N approaches the edge of phase space, the helicity flip is more significant and right-handed HNLs cannot be ignored.

Here we assume $E_\pi \gg m_\pi$ to simplify the boost process. This resembles the SM neutrino case,[21] with modifications arising from HNL mass and helicity separation.

3.2. *Production in muon decay*

Muons can decay into HNLs via a process analogous to their standard decay into ν_μ. The decay width of $\mu^- \to Ne^-\bar{\nu}_e$ has the same form as that of the previously

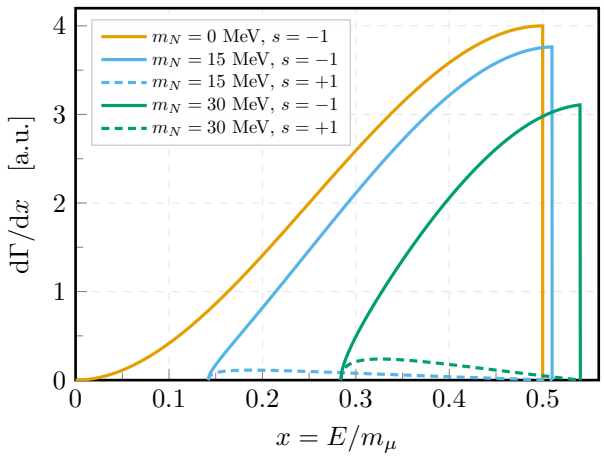


Fig. 4. Energy spectrum of HNL when muons decay into HNLs in the muon rest frame. The results are normalized to the $m_N = 0$ case, which corresponds to the SM neutrino and has no right-handed component. As m_N increases, the shape of the spectrum remains roughly the same, but the maximum energy increases while the total decay width decreases, yielding a increasing fraction of right-handed HNLs.

studied τ decay process[23]

$$\frac{\Gamma_{\mu\to N}}{\Gamma_{\mu\to\nu_\mu}} = |U_{\mu N}|^2 \times \rho_\mu(m_N) \tag{12}$$

where

$$\rho_\mu(m_N) = -r_N^8 + 8r_N^6 - 24r_N^4 \ln(r_N) - 8r_N^2 + 1, \quad r_N = m_N/m_\mu \tag{13}$$

The helicity-dependent spectrum of the HNL in the muon rest frame is given by

$$\begin{aligned}\frac{d\Gamma}{dE} = \frac{G_F^2}{24\pi^3}|U_{\mu N}|^2 \times |\mathbf{p}| \cdot [3(m_\mu^2 + m_N^2)E - 4m_\mu E^2 - 2m_N^2 m_\mu \\ - s \cdot |\mathbf{p}|(m_N^2 + 3m_\mu^2 - 4m_\mu E)]\end{aligned} \tag{14}$$

Several examples are plotted in Fig. (4).

A fully spin-correlated treatment could retain the polarization of the parent muons and the angular relation between their spin and decay products. Here we neglect parent-muon polarization as a simplifying approximation. This approximation isolates the HNL mass and polarization dependence but should not be interpreted as a demonstrated consequence of complete magnetic depolarization in the source. For a given muon spectrum, we have

$$\begin{aligned}Q_N^{(s)}(E_N) = \int_{\frac{m_N^2}{m_\mu^2}}^1 \frac{dy}{2y}\, Q_\mu(\frac{E_N}{y})|U_{\mu N}|^2 \rho_\mu \cdot \int_{x^-}^{x^+} \frac{dx}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \\ \cdot \left(\frac{1+\cos\theta_r}{2} g_N^{(s)}(x) + \frac{1-\cos\theta_r}{2} g_N^{(-s)}(x)\right)\end{aligned} \tag{15}$$

where

$$x^+ = \frac{m_N^2 + m_\mu^2}{2m_\mu^2},\ x^- = \frac{1}{2}\left(y + \frac{1}{y}\cdot\frac{m_N^2}{m_\mu^2}\right) \tag{16}$$

$$\cos\theta_0 = \frac{y - x}{\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}}} \tag{17}$$

$$\cos\theta_r = \frac{1}{y\sqrt{1 - \frac{m_N^2}{E_N^2}}}\left(\sqrt{x^2 - \frac{m_N^2}{m_\mu^2}} + x\cos\theta_0\right) \tag{18}$$

where $g_N^{(s)}(x)$ is normalized as

$$\int_{\frac{m_N}{m_\mu}}^{x^+} dx \sum_{s=\pm 1} g_N^{(s)}(x) = 1 \tag{19}$$

3.3. *Visible three-body decay*

Kinematically, low-mass HNLs in the ISS model can decay only into leptons. The possible final states are $\nu_\mu e^- e^+$ and $\nu_\mu \nu_l \bar{\nu}_l$. These processes have been well studied;[23–25] their decay widths are given by

$$\Gamma_{N\to 3\nu} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \tag{20}$$

$$\Gamma_{N\to \nu e\bar{e}} = \frac{G_F^2 m_N^5}{192\pi^3} \times |U_{\mu N}|^2 \times C_1^e \tag{21}$$

where

$$C_1^e = \frac{1}{4}\left(1 - 4\sin^2\theta_W + 8\sin^4\theta_W\right) \tag{22}$$

To simplify the problem, we neglect the electron mass. The 3ν final state includes all neutrino flavor combinations.

An HNL produced in the central region can affect the electromagnetic signal only if its energy is ultimately transferred to visible charged particles, thus the branching ratio for an HNL to undergo a visible decay is

$$BR_{N\to \nu e\bar{e}} = \frac{C_1^e}{1 + C_1^e} \approx 0.11 \tag{23}$$

This implies a substantial energy loss, since only about one-tenth of the HNL decays produce charged leptons.

The angular and energy distributions of electrons/positrons are

$$\begin{aligned} g_e^{(1)}(x,\cos\theta) =& \frac{8}{C_1^e}\cdot x^2\cdot\{[(4-16s_W^2+64s_W^4)x-(1-4s_W^2+28s_W^4)]\cos\theta \\ & -(4-16s_W^2+64s_W^4)x+(3-12s_W^2+36s_W^4)\} \end{aligned} \tag{24}$$

$$\begin{aligned} g_e^{(2)}(x,\cos\theta) =& \frac{16}{C_1^e}\cdot x^2\cdot\{[(6-24s_W^2+32s_W^4)x-(3-12s_W^2+14s_W^4)]\cos\theta \\ & -(6-24s_W^2+32s_W^4)x+(3-12s_W^2+18s_W^4)\} \end{aligned} \tag{25}$$

Here, $g_e^{(1)}$ and $g_e^{(2)}$ represent the differential distributions for the outgoing electron (e^-) and positron (e^+), respectively, with $x = E_e/m_N$ and $s_W = \sin\theta_W$. The angle θ is defined between the outgoing lepton momentum and the parent HNL spin. To obtain the total injected lepton spectrum, we sum over both e^- and e^+:

$$\begin{aligned} Q_e^{(Lab)}(E) = BR_{N\to\nu e\bar{e}}\cdot\int_E^{+\infty} dE_N\, Q_N^{(s)}(E_N) \\ \cdot\int_{\frac{E}{2E_N}}^{\frac{1}{2}}\frac{dx}{xE_N}\sum_{i=1,2} g_e^{(i)}(x, s(\frac{E}{xE_N}-1)) \end{aligned} \tag{26}$$

Evaluating the angular and energy integrals yields

$$\begin{aligned} Q_e^{(Lab)}(E) =& BR_{N\to\nu e\bar{e}} \\ & \cdot\int_0^1\frac{dy}{3y}\,Q_N^{(s)}(\frac{E}{y})[16y^3-27y^2+11-s\cdot(32y^3-63y^2+36y-5)] \end{aligned} \tag{27}$$

where $y = E/E_N$. The dependence on the Weinberg angle θ_W cancels in the sum: in the massless limit $m_e \to 0$, helicity-flipping mass terms vanish, and the vector–axial-vector (V–A) interference terms are odd under charge conjugation ($e^- \leftrightarrow e^+$) and therefore cancel between the electron and positron yields, leaving the total spectral shape independent of $\sin^2\theta_W$.

4. HNL in NGC 1068

The observed neutrino flux of NGC 1068 is commonly modeled as originating from charged-pion decay.[4] Combining the analytical neutrino spectrum from pion decays[21] with theoretical models of NGC 1068,[4] we adopt the following parameterization for the pion spectrum:

$$\Phi_{\pi^\pm}(E) = \Phi_0\cdot\left(\frac{E}{1\,\mathrm{GeV}}\right)^{-\Gamma}\cdot e^{-(E/E_c)^\beta} \tag{28}$$

with best-fit values

$$\Phi_0 = 4.7\times10^{-12}\,\mathrm{GeV}^{-1}\,\mathrm{cm}^{-2}\,\mathrm{s}^{-1}, \Gamma = 0.1, E_c = 6.2\,\mathrm{GeV}, \beta = 0.3 \tag{29}$$

visualized in visualized in Fig. (5). The parameters are obtained by fitting this form to the theoretical neutrino flux of Ref.,[4] since the uncertainties and limited energy

range of the IceCube data do not by themselves determine the pion spectrum. This spectrum provides the parent-particle input for the production calculation of Sec. 3.

4.1. *Decay length and the escape assumption*

For HNLs to mediate energy transport from the opaque core to a transparent external environment, their decay length must be sufficiently large. We have

$$l_{dec} = \frac{E_N}{m_N}\, c\, \frac{\hbar}{\Gamma_{tot}} = \frac{192\pi^3 c\hbar}{G_F^2(1+C_1^e)} \cdot \frac{E_N}{m_N^6 |U_{\mu N}|^2} \tag{30}$$

For example, setting $|U_{\mu N}|^2 = 10^{-5}$, $m_N = 20\,\mathrm{MeV}$, and $E_N = 1\,\mathrm{TeV}$ yields $l_{dec} \approx 0.4\,\mathrm{pc}$, which implies that HNLs are unlikely to escape into interstellar space, while some decays will still occur inside the optically opaque region.

Because $l_{dec} \propto E_N/(m_N^6 |U_{\mu N}|^2)$, the decay length is extremely sensitive to m_N: over the 10 MeV to 30 MeV range it varies by more than two orders of magnitude. Since $l_{dec} \propto 1/|U_{\mu N}|^2$, at the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative bound over the 10 MeV to 30 MeV range) the mixing is one order of magnitude larger than the value used above, so that l_{dec} is correspondingly shorter by about one order of magnitude (approximately 0.04 pc). Most HNLs then can decay inside the opaque core and their visible products are absorbed before escape. We therefore adopt the optimistic assumption that all HNLs decay beyond the core and that their energy escapes unattenuated; this assumption maximizes the predicted flux and makes the resulting exclusion conservative.

4.2. *Environment and radiation calculation*

We refer to the parameters of the AGN torus model.[26] For our approximation we set the magnetic field strength to 139 mG, dust temperature to 1500 K, ambient density to $2.30 \times 10^5\,\mathrm{cm}^{-3}$, and bolometric luminosity to $9.55 \times 10^{44}\,\mathrm{erg\,s}^{-1}$, and assume that electron-photon conversion happens at about 0.4 pc from the AGN core, corresponding to the inner edge of the torus in this model.

The electron/positron source function $Q_e^{(Lab)}(E)$ derived in Sec. 3.3 is injected into GAMERA,[11] which computes synchrotron, inverse-Compton (on the dust and CMB photon fields), and bremsstrahlung emission from these leptons in the adopted environment. Internal $\gamma\gamma$ absorption is neglected as an optimistic assumption; including it would further reduce the predicted flux. The resulting photon spectrum is shown in Fig. (7).

4.3. *Comparison with observations and the suppression budget*

Setting $|U_{\mu N}|^2 = 1$, the predicted photon flux lies three to four orders of magnitude below the observational upper limits. The residual suppression arises mainly from the small visible branching ratio $BR \simeq 0.11$, which contributes about one order of magnitude, and from the fraction of the parent pion or muon energy carried into the

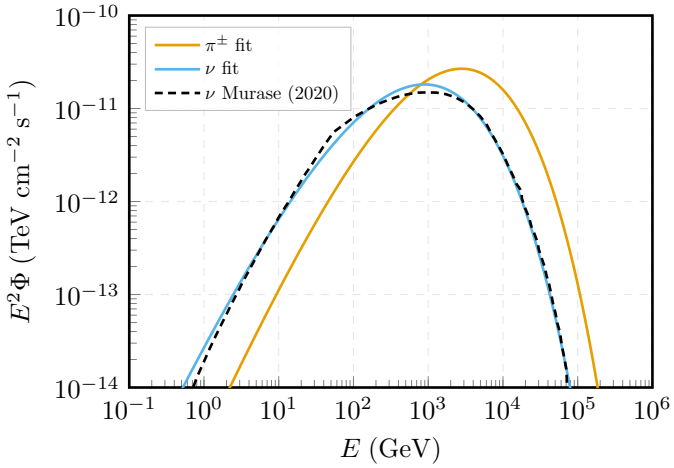


Fig. 5. Parametrization of the pion spectrum in NGC 1068 (Eq. (28)) for theoretical prediction.[4]

visible channel together with the radiation efficiency of the adopted environment, which account for the remaining orders. The phase-space factors ρ_π, ρ_μ are of order unity over most of the 10 MeV to 30 MeV range and differ substantially from unity only near the kinematic endpoint, so they are not counted as a source of orders of magnitude. Applying the experimental bound $|U_{\mu N}|^2 \lesssim 10^{-4}$ (the representative value over this range) adds four more orders of magnitude, bringing the total suppression to seven to eight orders.

Two qualifications do not alter this result. First, the experimental bound on $|U_{\mu N}|^2$ is mass dependent and changing over the entire 10 MeV to 30 MeV range, although the difference is minor. Second, an accurate flux from a finite external region must account for the probability of decay within that region; the present treatment instead assumes that all HNLs deposit their visible products in the chosen zone, which makes the estimate optimistic rather than fully propagated. The first qualification affects the bound only at the same order, and the second can only reduce the predicted flux, so the exclusion is conservative. The comparison in Fig. (7) uses the Fermi-LAT point-source upper limits.[3] Within the range $m_N \simeq 10$ MeV to 30 MeV, the channel is therefore excluded as an explanation of the γ-ray excess, although the laboratory limits on $|U_{\mu N}|^2$ are unchanged.

5. Conclusions

We have presented a polarization-resolved calculation of HNL production and decay in the muon-neutrino sector and applied it to the multi-messenger observations of NGC 1068. The HNL spectra from $\pi^{\pm} \to N + \mu^{\pm}$ and $\mu^{\pm} \to N + e^{\pm} + \bar{\nu}_e$ were computed separately for the two polarization states, boosted to the laboratory frame with the mass-dependent polarization redistribution retained, and converted into the electron/positron spectrum of the visible decay $N \to \nu_\mu + e^+e^-$. The resulting

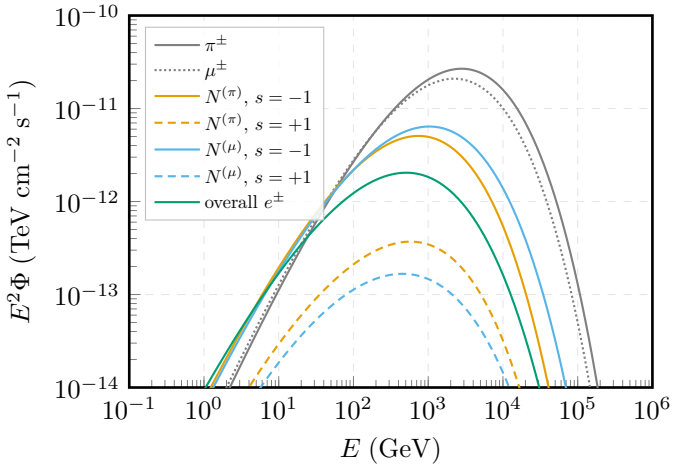


Fig. 6. HNL production source function from pion and muon decay for NGC 1068, using Eq. (28) as source pion spectrum. The polarization of muon is omitted. HNL mass is set to $m_N = 10\,\mathrm{MeV}$, and all HNL spectrums are scaled to $|U_{\mu N}|^2 = 1$ to better show the physics. For $m_N = 10\,\mathrm{MeV}$, right-handed neutrino flux is 1 to 2 orders of magnitude smaller than left-handed one.

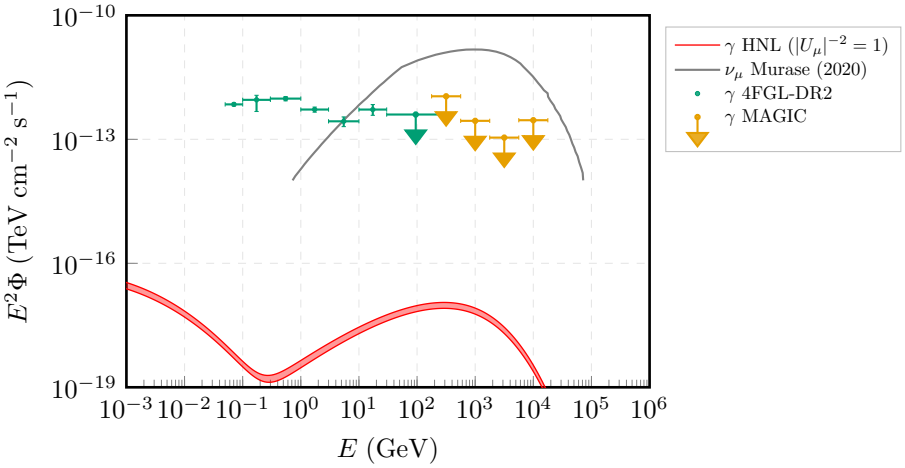


Fig. 7. Possible HNL generated photon spectrum compared with observations. $m_N = 10\,\mathrm{MeV}$ to $30\,\mathrm{MeV}$ is shown as a range. Even ignoring the mixing angle, which is smaller than 10^{-4}, the flux is 3 to 4 orders of magnitude smaller than the observational upper limit, which means it is at least 7 to 8 orders of magnitude smaller and therefore can be completely ignored in observation.

γ-ray emission, evaluated for a representative environment of the source, lies at least 7 to 8 orders of magnitude below the observational limits once the laboratory constraint on $|U_{\mu N}|^2$ is imposed, and remains 3 to 4 orders of magnitude below them even for $|U_{\mu N}|^2 = 1$. The weakly coupled HNL channel is therefore excluded

as the origin of the γ-ray excess. The suppression factorizes approximately into the visible branching ratio, the fraction of the parent energy transferred to the electron–positron pair, the radiative efficiency of the environment, and the mixing-angle constraint (Sec. 4.3).

The approximations adopted here are conservative—in particular, all HNLs are assumed to decay within the adopted emission region—so a more complete treatment of transport can only reduce the predicted flux. The mass dependence of the laboratory bound on $|U_{\mu N}|^2$ over the 10 MeV to 30 MeV range is a minor residual uncertainty (Sec. 4.3). The present analysis does not modify the laboratory limits on $|U_{\mu N}|^2$, but it converts the multi-messenger data of NGC 1068 into an exclusionary constraint on this HNL channel that is complementary to laboratory searches.

The negative result is itself informative. It excludes a well-motivated weakly coupled transport channel for the excess and shifts the emphasis to alternatives, most notably a refined description of the AGN environment underlying current electromagnetic cascade models,[4] or energy transport mediated by other weakly interacting particles. The polarization-resolved framework is not specific to this source or mass window: it applies to any hadronic source in which HNLs are produced, to other HNL masses, and to other neutral particles whose visible decay products retain polarization information.

Finally, the helicity-tracking method is directly relevant to accelerator-based HNL analyses. Polarization effects have already been included in accelerator simulations for the DUNE near detector.[10] Polarization-aware decay distributions could similarly be incorporated into the displaced-vertex and track reconstruction of SHiP,[27] the LArTPC mass reconstruction,[28] and the associated acceptance and likelihood calculations. Such inputs could also be used to reinterpret null searches and optimize detector geometries.

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